



04.18 F: What is deep learning?

Completion requirements

- Topics: **What is deep learning?**
 - Perceptrons and Artificial neural nets (ANN) revisited
 - Deep learning (DL)
 - Basic network architecture:
 - Input layer
 - Output layer
 - Hidden layers
 - Activation function
 - Tools: tensorflow, keras, GPUs
- Reading:
 - [A Gentle Introduction to Neural Networks \(Towards Data Science\)](#)
 - [Tutorial: Your First Deep Learning Project in Python with Keras Step-By-Step \(Machine Learning Mastery\)](#)
- Additional Resources (optional, but encouraged):
 - [The Math Behind Neural Networks \(SkyMind\)](#) [A little mathy, but you would expect that from the title!]
 - [Deep Feedforward Networks](#) [Note: This is a more rigorous explanation of neural networks. It is especially recommended for DSC majors who have taken Calc I-III and Linear Algebra. It is a little more technical than what we are discussing in DSC 305, but more on the level of what you would find in a graduate ML course. Incidentally, this reading is on my syllabus draft for DSC 395.]

Last modified: Friday, April 18, 2025, 1:39 PM